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Nance-Horan Syndrome: Localization within the Region Xp21.1-Xp22.3 by Linkage Analysis

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Summary

Nance-Horan Syndrome (NHS) or X-linked cataract-dental syndrome (MIM 302350) is a disease of unknown pathogenesis characterized by congenital cataracts and dental anomalies. We performed linkage analysis in three kindreds with NHS by using six RFLP markers between Xp11.3 and Xp22.3. Close linkage was found between NHS and polymorphic loci DXS43 (θ = 0 with lod score 2.89), DXS41 (θ = 0 with lod score 3.44), and DXS67 (θ = 0 with lod score 2.74), defined by probes pD2, p99-6, and pB24, respectively. Recombinations were found with the marker loci DXS84 (θ = .04 with lod score 4.13), DXS143 (θ = .06 with lod score 3.11) and DXS7 (θ = .09 with lod score 1.68). Multipoint linkage analysis determined the NHS locus to be linked completely to DXS41 (lod score = 7.07). Our linkage results, combined with analysis of Xp interstitial deletions, suggest that the NHS locus is located within or close to the Xp22.1-Xp22.2 region.

Introduction

The Nance-Horan Syndrome (NHS), or cataract-dental syndrome (McKusick 302350), is an X-linked hereditary disease characterized by dental anomalies and cataracts (Horan and Billson 1974; Nance et al. 1974). The dental anomalies in affected males are characterized by screwdriver-shaped incisors, wide spacing of teeth (diastema), and supernumerary maxillary incisors. The dental anomalies in female carriers are similar but less severe. The cataracts in affected males involve the fetal nucleus, the posterior Y-suture, and variable zonular extensions into the posterior cortex. Carrier females have posterior Y-sutural opacities of the lens. Other features such as anteverted pinnae, short fourth metacarpal bones, and microcornea are phenotypic variables which may be absent in affected males (van Dorp and Delleman 1979; Bixler et al. 1984).

NHS has been classified separately from X-linked cataracts (XLC) without dental anomalies (McKusick 302200). Although the cataract phenotype is similar in both disorders, the absence of other reported manifestations in XLC has resulted in a separate classification. Because both the genetic loci and the pathogenesis of both NHS and XLC are unknown, it is unclear whether these are allelic disorders, part of a contiguous gene deletion syndrome, or the result of mutations at entirely separate loci. The XLC locus has been loosely linked to the Xg blood group at a lod score (Z) of <1.0 (Fraccaro et al. 1967). There are no data mapping the NHS locus.

The aim of the present study was to map the NHS gene by RFLPs and by DNA-based linkage analysis in order to establish the most closely linked polymorphic markers, as a first step toward characterizing the molecular basis of NHS and XLC. We report here the results of linkage analysis in four NHS families with six X-chromosomal loci that span the region between Xp21 and Xp22.

Material and Methods

Family Studies

Four families in which NHS is segregating were ascertained and examined by one of us (R.A.L.). Detailed

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Table 1

RFLP Loci and Probes Used between Xp21 and X pter

Locus	Probe	Location	Enzyme	Alleles	Frequencies
DXS143	dic56	Xp22.3-22.2	*Bcl*I	7.4 and 8.9	.56 and .44
DXS9	RC8	Xp22.3-Xp22.2	*Taq*I	3.2 and 3.0	.84 and .06
DXS43	pD2	Xp22.2	*Pvu*II	6.0 and 6.6	.55 and .45
DXS41	p99-6	Xp22.1	*Pst*I	22.0 and 13.0	.71 and .29
DXS67	pB24	Xp21.3	*Bst*N1	1.6 and 1.4	.85 and .15
DXS84	L754	Xp21.1	*Pst*I	12.0 and 9.0	.62 and .38
DXS7	L1.28	Xp11.3	*Taq*I	12.0 and 9.0	.68 and .32

family histories and pedigrees were obtained through personal interviews with family members in each kindred. Available family members underwent careful ocular examinations including slit-lamp and fundus examination and measurement of horizontal corneal diameters. Nonocular manifestations such as abnormal teeth and shortened fourth metacarpals were noted. Females at risk were scored as NHS heterozygotes if they had ophthalmologic evidence of characteristic lenticular opacities involving, at a minimum, the posterior Y-suture. The cataracts were characteristically fine, punctate opacities aligning with or outlining the posterior Y-suture. In some instances, the carrier status of females was inferred if the pedigree structure showed them to be obligate heterozygotes. The criteria used for identifying affected males included the presence, on biomicroscopy, of central and coarse posterior Y-sutural cataracts, microcornea (horizontal diameter ≤11 mm in an otherwise structurally normal cornea), and dental abnormalities.

DNA Analysis

Peripheral venous blood was collected in EDTA tubes and used as a source of leukocyte DNA. Heparinized blood was also collected for establishing permanent lymphoblastoid lines by Epstein-Barr virus transformation (Wilson et al. 1982). Genomic DNA from all available family members was prepared with an automated DNA extractor (Applied Biosystems, Foster City, CA) and was digested and blotted by standard techniques (Southern 1975).

The DNA probes used in this study are listed in table 1. Purified probes were radiolabeled by random priming (Feinberg and Vogelstein 1983) and hybridized to Southern blots by standard techniques.

Linkage Analysis

The seven DNA marker loci used were DXS143 (Drayna and White 1985; Middlesworth et al. 1985), DXS9 (Hartley et al. 1984; Wieacker et al. 1984), DXS43, DXS41, and DXS67 (Aldridge et al. 1984; de Martinville et al. 1985), and DXS84 (Hofker et al. 1986). The polymorphic probes used to detect the RFLP and their location are listed in table 1.

Linkage analysis was performed with the Z method (Morton 1955). Each marker locus was a codominant,

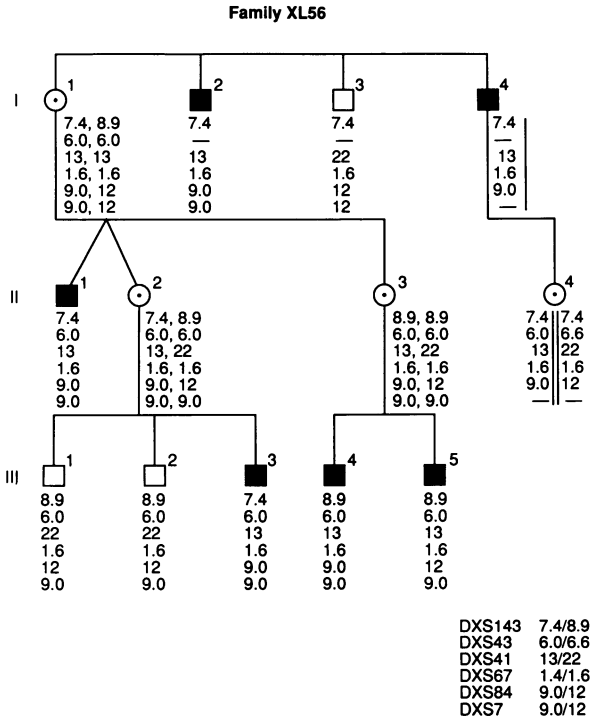


Figure 1 Segregation of RFLP alleles with NHS in family XL56. The summary of the probes and alleles is given below in the same order as that for the data presented below each pedigree symbol. When phase is known, alleles in coupling are placed on the same side of a vertical line. The mother II-3 is heterozygous for DXS84. Although phase is unknown, one of her son's, III-4 or III-5, is a recombinant for DXS84. The mother I-1 is heterozygous for DXS143. Both her daughters, II-2 and II-3, have been scored as carriers, and either one could represent a recombinant for DXS143.

biallelic system. Published allele frequencies for Caucasian populations were used for each marker. Standard Z tables for each marker locus and for NHS were obtained with the LINKAGE program MLINK (version 4.9) (Lathrop et al. 1985). Maximum-likelihood estimates of recombination ($\hat{\theta}$) were obtained with ILINK. Significance was evaluated using the standard criterion: odds favoring linkage must be $\geq$1,000:1 to establish linkage ($Z \geq 3.0$).

Multipoint linkage analysis was performed with NHS and markers in the chromosomal region showing significant linkage with the disease locus. The localization within the multipoint map was performed by the location-score method (Cook et al. 1974; Lathrop et al. 1985) as implemented in the LINKAGE program LINKMAP. Specifically, the likelihood of the NHS locus was evaluated in all possible locations on a fixed multilocus map. A multilocus Z was calculated for each possible location (on the assumption that there was no interference) by contrasting the likelihood of this position with the likelihood that the NHS locus was unlinked to the multilocus map. An interval was excluded as containing the NHS locus if its relative likelihood was 1,000 times less likely than the best (maximum-likelihood) location. A 1-Z difference from the maximum-likelihood estimate was used to construct a support interval (Conneally et al. 1985).

Results

The results of the RFLP analysis are summarized in the three pedigrees (shown in figures 1–3). The fourth pedigree is not included in the analysis since the females were uninformative for the seven probes listed in table

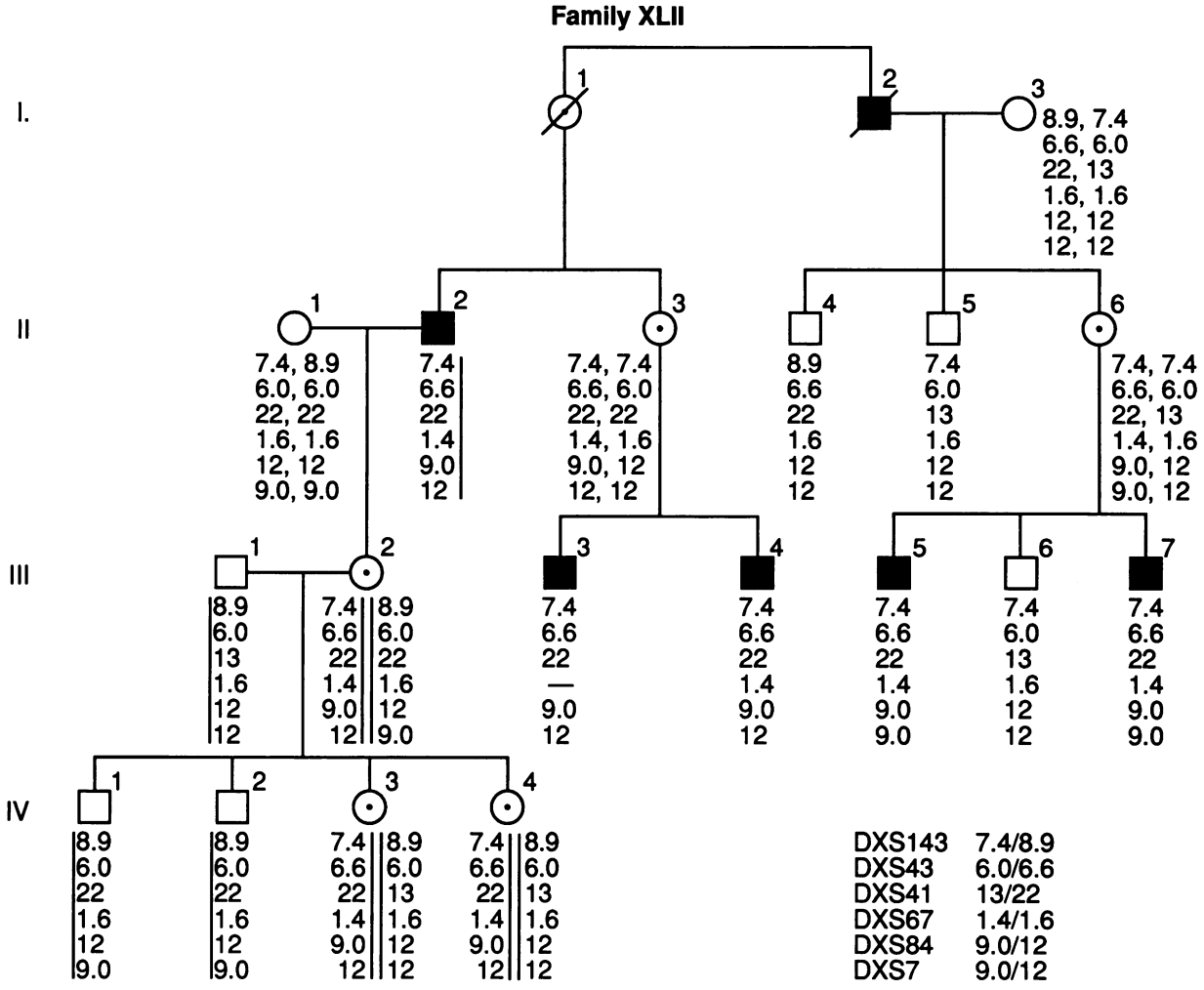


Figure 2 Segregation of RFLP alleles with NHS in family XL11

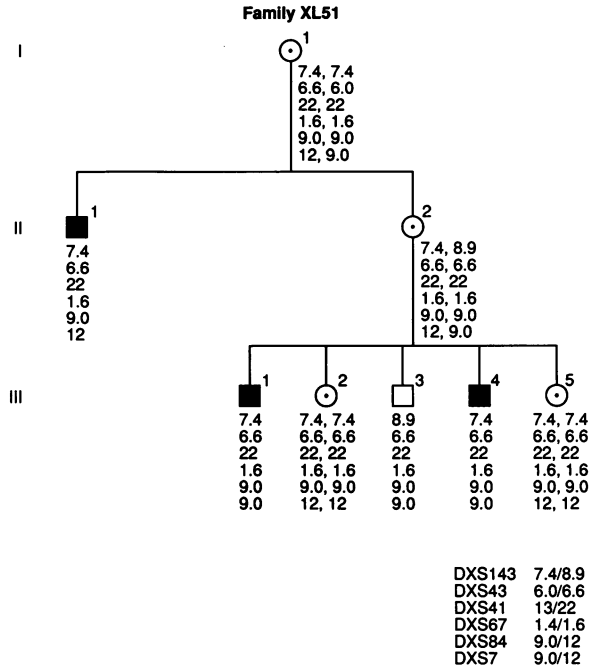


Figure 3 Segregation of RFLP alleles with NHS in family XL51. The mother II-2 is heterozygous for DXS7. Although phase is unknown, one or more of her son's, III-1, III-3, or III-4, is a recombinant for DXS7.

1. The order of the loci in each haplotype follows the currently accepted order of the loci on the short arm of the X chromosome: centromere–L1.28(DXS7)–L754(DXS84)–pB24(DXS67)–p99–6(DXS41)–pD2(DXS43)–RC8(DXS9)–dic56(DXS143)–telomere (Mandel et al. 1989). Genetic distances in centimorgans (cM) were assumed to be as follows: DXS143–13.8–DXS43–13.7–DXS41–21.1–DXS84–16.5–DXS7 (C. B. Keats [Human Gene Mapping Committee on Linkage], personal communication). The results of the two-point analysis are shown in table 2. The marker DXS9 (RC8) was not included in the two-point linkage analysis since it was uninformative for most carrier females tested. For each RFLP locus, values for $\hat{\theta}$ and maximum Z (Z_{max}) are given. On the basis of its ability to detect carrier females clinically, NHS was modeled as an X-linked locus with 98% penetrance in carrier females. This penetrance was assumed because no obligate female carriers in our families lacked cataract changes. When the estimate of penetrance was varied from 95% to 100%, the Z values were not altered substantially. The frequency of the disease allele in the population was assumed to be .0001. The Z values were also calculated with a mutation rate of 10^{-5}, without significant change in Z values.

In all three scored families, NHS was found to be completely linked to RFLP loci DXS43, DXS41, and DXS67. DXS84 and DXS143 also demonstrated close linkage with recombination frequencies (θ) of .04 (Z = 4.13) and .06 (Z = 3.11), respectively.

Multipoint linkage analysis was used to determine the best location for NHS in this region of the X chromosome. Location-score analysis was performed with a map of marker loci previously localized to this region. The genetic distances for this map were obtained from B. Keats (personal communication). Existing multipoint maps of the X chromosome do not provide a unique location for DXS67. Therefore, this locus was not included in the location-score analysis. The current data set did not contain sufficient information to uniquely place DXS67 or revise this map (data not shown). The location-score analysis is summarized in figure 4. The best location for NHS was observed to be completely linked to DXS41 (multilocus Z = 7.07). The 1-Z support units placed NHS in the interval 0.06 distal to DXS84 and 0.05 proximal to DXS143. NHS was excluded, with odds of >1,000:1, from lying in the region distal to DXS143 and proximal to DXS7. NHS could not be excluded, by these criteria, from other intervals within the anchor map.

Table 2

Results of Pairwise Linkage Analysis between NHS and Xp Markers

	Z *at* θ OF								
LOCUS	.0	.001	.05	.1	.2	.3	.4	θ	Z_{max}
DXS143	∞	1.75	3.11	3.03	2.54	1.85	1.00	.06	3.11
DXS43	2.89	2.88	2.60	2.31	1.71	1.10	.51	.00	2.89
DXS41	3.44	3.44	3.14	2.82	2.16	1.43	.67	.00	3.44
DXS67	2.74	2.74	2.49	2.23	1.68	1.01	.50	.00	2.74
DXS84	∞	2.91	4.12	3.90	3.09	2.05	.95	.04	4.13
DXS7	∞	1.35	1.61	1.68	1.45	1.04	.54	.09	1.68

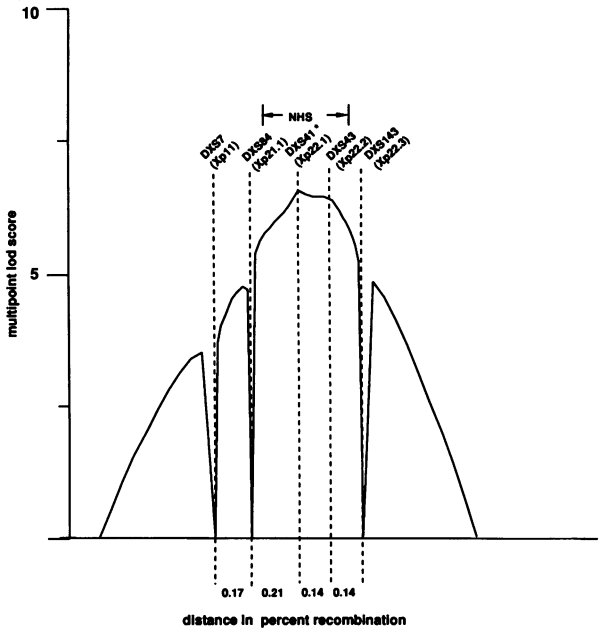


Figure 4 Summary of NHS location-score analysis. Shown here is the multilocus *Z* corresponding to the placement of NHS in each interlocus interval. The numbers presented for each interval indicate percent recombination between points in the map. The bracket surrounding NHS indicates the 1-*Z* support interval. The star next to the DXS41 locus represents the best location for NHS within the map. DXS67 was not included in the multipoint analysis, since the physical distance between this locus and surrounding loci is not known.

Discussion

We have used linkage analysis in three informative families to show that the gene responsible for NHS is on the short arm of the X chromosome in the region from Xp21.1 to Xp22.3. The locus for NHS appears to sit between the flanking markers DXS84 and DXS143 and shows tight linkage to DXS43, DXS41, and DXS67. Although multipoint analysis locates NHS at or near DXS41, with a *Z* of 7.07, the position of the locus with respect to DXS67, DXS41, and DXS43 cannot be unequivocally determined, because of the absence of recombination events between NHS and these three loci in our families.

The regional localization for NHS can be further narrowed, however, using published data from patients with interstitial deletion syndromes involving the short arm of the X chromosome. Clarke et al. (1986) have reported a boy with Duchenne muscular dystrophy, adrenal hypoplasia, glycerol kinase deficiency, and mental retardation associated with an interstitial deletion of the short arm of the X chromosome, from Xp21.1 to Xp22.1. Ocular examination revealed no pathology. Analysis of the cytogenetic deletion with X-chromosome short-arm probes revealed that DXS84, DXS164, and DXS28 were missing but that the DXS41 locus was intact, localizing the breakpoint proximal to DXS41. Other case reports have described cytogenetic and probe deletions involving Xp21.1→Xp21.3, with no mention of any cataracts (Francke 1984; Francke et al. 1985; Greenberg et al. 1987; Marlhens et al. 1987; Chelly et al. 1988). These deletions, when considered in combination with the linkage data, narrow the location of the NHS locus within a bracketed region slightly proximal to DXS41 and to a point just proximal to DXS143.

Cataracts and dental anomalies in NHS may represent pleiotropic effects of a single gene defect, which could be due to many different kinds of mutations. However, the multisystem phenotype could also be the result of a contiguous gene syndrome (Schmickel 1986) due to a submicroscopic cytogenetic deletion. The recognizable phenotype of these syndromes results from deletion of several functionally unrelated genes physically contiguous on a chromosome. Such syndromes as Langer-Giedion (del 8q24), aniridia-Wilms tumor (del 11p13), Prader-Willi (del 15q11-12), DiGeorge (del 22q11), and choroideremia-deafness (Xq21) involve deletion of multiple unrelated loci (Emanuel 1988). Clinical variation within these syndromes may be caused by different deletions involving overlapping sets of genes. This provides a plausible model for a genetic basis of isolated XLC: cataracts and the other physical abnormalities of NHS (e.g., microcornea and dental and skeletal abnormalities) may result from deletion of independent but contiguous loci on Xp, whereas mutation or deletion of the putative cataract locus alone would be a possible etiology for some cases of isolated XLC. It is also possible that the several reports of isolated XLC (Walsh et al. 1937; Fraccaro et al. 1967; Krill et al. 1969) are misclassifications because other features of NHS were not documented. For example, one of us (R.A.L.) reexamined the family reported by Krill et al. (1969) and found microcornea and hand, ear, and dental anomalies consistent with the NHS. Therefore, the documentation of the other features of NHS may vary with the era in which the family was ascertained and with the clinical specialties of the principal authors.

The proximity of DXS41 to human X-linked hypophosphatemia (Machler et al. 1986; Read et al. 1986; Thakker et al. 1987) is interesting, since the mouse locus for hypophosphatemia is known to be 5 cM telomeric to the mouse XLC locus, Xcat (Favor and Pretsch 1987). The mouse Xcat mutation causes total lens opacity both in hemizygous males and in homozygous

females. Heterozygous females have a phenotype varying from totally opaque to clear lenses and including intermediate phenotypes, as expected for an X-linked gene that undergoes X inactivation. If it is assumed that the Xcat locus of mouse corresponds to the human NHS locus, our location is compatible with the scheme proposed by Buckle et al. (1984, 1985), who hypothesize that large segments on the mouse and human X chromosomes have been conserved during evolution. Therefore, the proximity of the mouse Xcat and hypophosphatemia loci, as well as the tight linkage of both the human hypophosphatemia and NHS locus to DXS41, may suggest a conserved block of genes in this region of the X chromosome.

In summary, our study of NHS in four families has identified several genetic markers closely linked to NHS and supports a map assignment of NHS to Xp21.1-Xp22.3. The additional information obtained from previous case reports with interstitial deletions supports a further narrowing of this locus assignment to Xp22.1-Xp22.3. The subregional mapping of NHS enables a concentrated search for deletions and closer genetic markers to be carried out within this chromosomal segment, in order to elucidate further the genetic abnormality causing this disorder.

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